Unit 7 e-Portfolio Activity: Perceptron Activities

Introduction

I ran the three supplied notebooks in order, corrected in six places, and added a check of my own after each exercise.

Method

The perceptron in Exercises 1 and 2 uses threshold 1 with no trainable bias. Its supplied update used the absolute error, so the weights could never fall after a false positive; I used the signed error of the perceptron learning rule (Géron, 2025), capped at 100 epochs and stopping earlier after a zero-error pass. In Exercise 3, I fixed an undefined variable that broke the training loop and seeded the starting weights.

Findings

The perceptron learnt AND in 6 epochs and OR in 10, but not XOR, which no linear function of the two inputs can produce (Goodfellow, Bengio and Courville, 2016). It also failed NAND, with 1 of 4 rows correct: NAND is linearly separable, but this fixed-threshold implementation always outputs 0 for (0, 0), whereas the lecturecast's neuron has a bias (University of Essex Online, 2026). With a trainable bias added, it learnt NAND in 6 epochs, while XOR still failed (Table 2). The multi-layer perceptron, also without biases, learnt XOR using the sigmoid, which Rumelhart, Hinton and Williams adopted so that backpropagation has a gradient to follow (1985, cited in Géron, 2025). Its error fell from 0.500 to 0.012 in 100,000 epochs, then only to 0.006 in 300,000 more (Figure 1); by the end, half its hidden activations and all its outputs sat near 0 or 1 (Table 4). Such saturation is why sigmoid hidden units are now discouraged (Goodfellow, Bengio and Courville, 2016). Five other starting points, at 100,000 epochs each, also learnt XOR: a small check, not proof of robustness.

Implications and conclusion

Whether a model can learn a dataset depends on the data's shape and the model's built-in assumptions, not only on training time. One minus the mean absolute error measures closeness to the targets, not the share classified correctly. Four rows of logic data leave nothing unseen, so these runs show learning, not generalisation.

Perceptron evidence

Weights	Weighted sum for inputs (45, 25)	Output
[0.7, 0.1] (supplied)	34.00	1
[−0.7, 0.1] (supplied)	−29.00	0
[0.02, 0.01]	1.15	1
[0.0, 0.04]	1.00	1

Table 1: Exercise 1. The step function fires at a weighted sum of 1 or more, so small weights can still fire.

Operator	One line separates it?	Threshold 1, no bias	Correct	Trainable bias	Correct
AND	Yes	Learnt, 6 epochs	4 of 4	Learnt, 4 epochs	4 of 4
OR	Yes	Learnt, 10 epochs	4 of 4	Learnt, 4 epochs	4 of 4
NAND	Yes	Not learnt, 100 epochs	1 of 4	Learnt, 6 epochs	4 of 4
XOR	No	Not learnt, 100 epochs	2 of 4	Not learnt, 100 epochs	2 of 4

Table 2: The Exercise 2 learning rule (signed error, learning rate 0.1, capped at 100 epochs, stopping after a zero-error pass) on four operators: as supplied, with threshold 1 and no trainable bias, then with a trainable bias and threshold 0. AND as supplied is the exercise; the rest is my check.

Multi-layer perceptron evidence

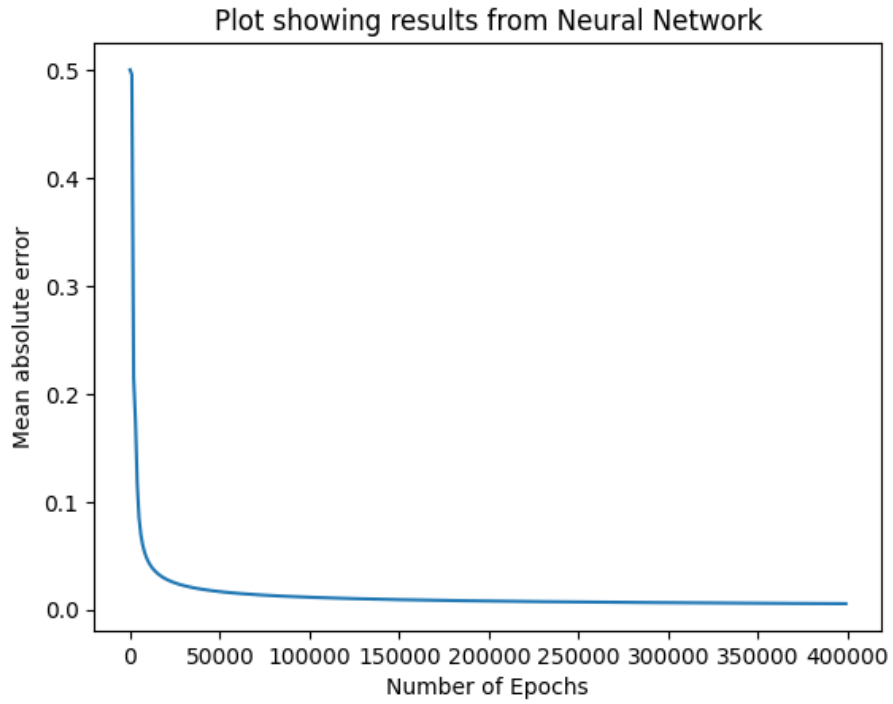


Figure 1: Mean absolute error of the Exercise 3 network on XOR, recorded every 1,000 epochs (learning rate 0.6, seed 42).

Starting weights	Epochs	Final error	Outputs for (0,0), (0,1), (1,0), (1,1)	Correct
Seed 42 (main run)	400,000	0.0057	0.004, 0.993, 0.994, 0.006	4 of 4
Seed 0	100,000	0.0173	0.022, 0.984, 0.984, 0.015	4 of 4
Seed 1	100,000	0.0076	0.013, 0.992, 0.992, 0.002	4 of 4
Seed 2	100,000	0.0114	0.009, 0.988, 0.986, 0.011	4 of 4
Seed 3	100,000	0.0188	0.020, 0.984, 0.984, 0.023	4 of 4
Seed 4	100,000	0.0169	0.021, 0.985, 0.984, 0.015	4 of 4

Table 3: The XOR network from different starting weights. The target outputs are 0, 1, 1, 0; ‘correct’ means the rounded output matches. The final error is the mean absolute difference from the targets, computed from the final weights.

Measure, after 400,000 epochs (seed 42)	Value
Mean absolute error at epoch 100,001 and at the end	0.0116 and 0.0057
Largest slope of an output unit (maximum possible 0.25)	0.0068
Hidden activations below 0.05 or above 0.95	6 of 12
Largest absolute weight	28.5

Table 4: Saturation at the end of training. A sigmoid slope near zero can attenuate gradients during backpropagation.

Corrections to the supplied notebooks

Exercise	Problem	Correction
2	The weight update used the absolute error, so it could only raise the weights.	Signed error.
2	Training ran until the error was zero, so it never stops on data it cannot learn.	Stop after 100 epochs.
3	A cell raises a shape error on purpose, which stops ‘Run all’.	The error message is printed instead.
3	The training loop used weights1, which is never defined, so it failed on a fresh run.	weights_1.
3	Random starting weights without a seed.	Fixed seed (42).
3	The error plot had four points but an axis in epochs; two cells used numbers from another run; one plot cell was duplicated.	Error recorded every 1,000 epochs and plotted against the epoch; this run’s numbers used; duplicate removed.

Table 5: The six corrections, also listed in the notebook’s opening cell.

References

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. Cambridge, MA: MIT Press. Available at: https://www.deeplearningbook.org/ (Accessed: 2 November 2026).

University of Essex Online (2026) 'Unit 7 Lecturecast: Artificial Neural Network (ANN)' [Lecturecast]. ML_PCOM7E: Machine Learning October 2026. University of Essex Online. Available at: https://www.my-course.co.uk/pluginfile.php/1659716/mod_scorm/content/2/scormcontent/index.html (Accessed: 25 October 2026).

Supplied materials: University of Essex Unit 7 e-Portfolio brief and the three notebooks by Dr Mike Lakoju, Cardiff Metropolitan University: simple_perceptron.ipynb, perceptron_AND_operator.ipynb and multi-layer Perceptron.ipynb.

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